CABIN AIR FILTER

Cabin Air Filter - Edge, MKX

REMOVAL & INSTALLATION

NOTE: Manufacturer's terminology for this filter is Cabin Air Filter, or Air Inlet Screen.

A cabin air filter is available as standard equipment on dual-zone EATC vehicles. EMTC vehicles are not equipped from the factory with a cabin air filter, but are equipped with an air inlet screen in place of the filter to prevent any large foreign material from entering the heater core and evaporator core housing. The cabin air filter or screen can be accessed by removing the glove compartment and removing the cabin air filter access cover.

A cabin air filter can be installed as a dealer accessory on vehicles not originally equipped with a filter.

CAUTION: Only a screen or a filter should be installed, never both.

1. The cabin air filter is located behind glove box. Remove glove box.
2. Remove filter cover (7). See **Fig. 1**
3. Remove cabin air filter (6) See **Fig. 1**

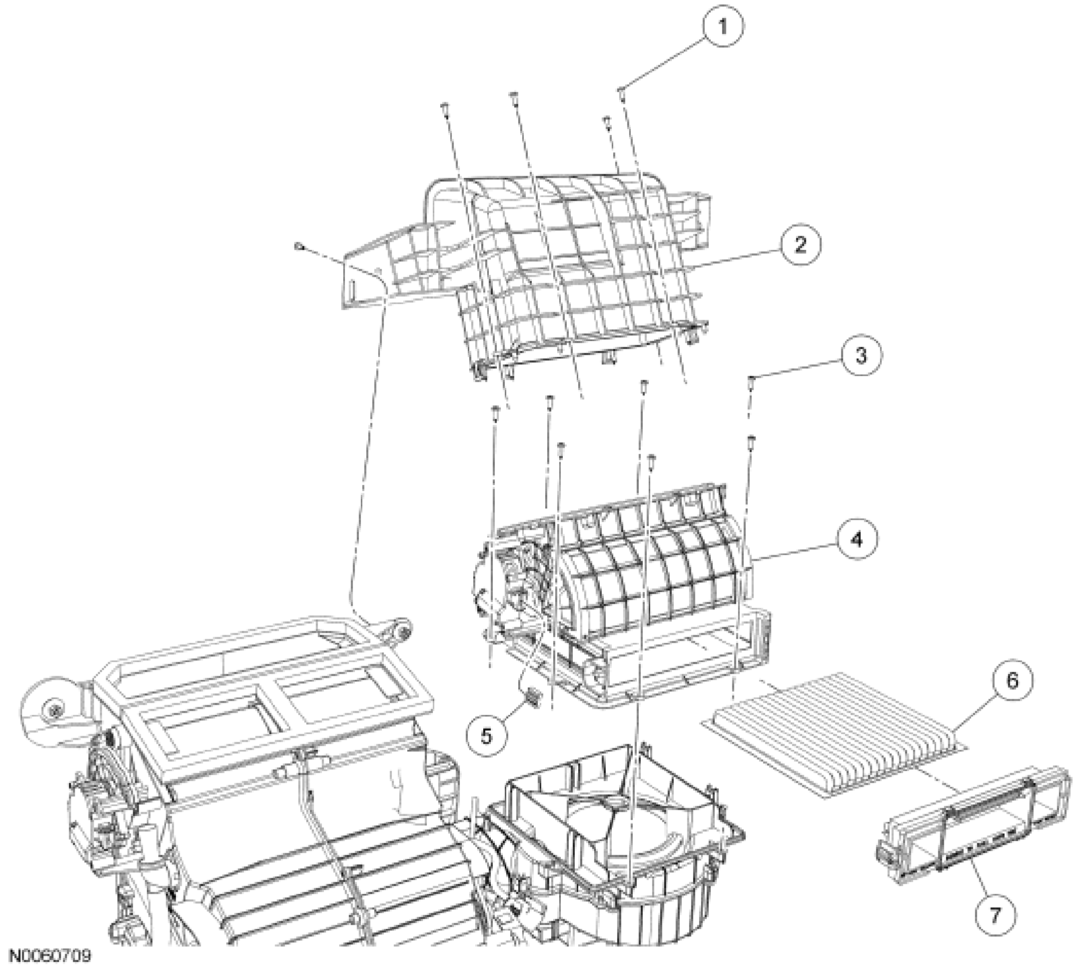


Fig. 1: Exploded View Of Air Inlet Duct
Courtesy of FORD MOTOR CO.

2008 GENERAL INFORMATION

Charging System - General Information - Edge & MKX

SPECIFICATIONS

GENERAL SPECIFICATIONS

Item	Specification
Generator	
Rating	78/143 amp (max) @ 620-2,500 engine RPM @ normal engine temperature
Voltage	12 volt

DESCRIPTION AND OPERATION

CHARGING SYSTEM

The charging system consists of the following:

- Generator
- Charging system warning indicator
- Necessary wiring and cables

The charging system is a negative ground system. The generator (including an internal voltage regulator) is belt-driven by the engine accessory drive system. When the engine is started, the generator begins to generate AC, which is internally converted to DC. This current is then supplied to the vehicle electrical system through the output (B+) terminal of the generator.

Battery

The battery is a 12V DC source connected in a negative ground system. The battery case is sealed with 2 vent holes to release gases. The battery has 3 major functions:

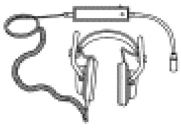
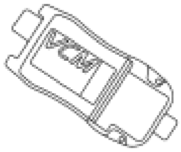
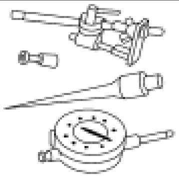
- Engine cranking power source
- Voltage stabilizer for the electrical system
- Temporary power source when electrical loads exceed the generator output current

DIAGNOSTIC TESTS

CHARGING SYSTEM

Special Tools

Illustration	Tool Name	Tool Number

 ST2311-A	73III Automotive Meter	105-R0057 or equivalent
 ST2312-A	Flex Probe Kit	105-R025B
 ST2834-A	SABRE Premium Battery and Electrical System Tester	010-00736 or equivalent
 ST1214-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	

Principles of Operation

The PCM controlled charging system determines the optimal voltage setpoint for the charging system and communicates this information to the voltage regulator. This system is unique in that it has 2 communication lines between the PCM and the generator/regulator. Both of these communication lines are pulse-width modulated (PWM). The generator communication (GENCOM) line communicates the desired setpoint from the PCM to the voltage regulator. The generator monitor (GENMON) line communicates the generator load and error conditions to the PCM. The third pin on the voltage regulator, the A circuit pin, is a dedicated battery voltage sense line.

The generator charges the battery and at the same time supplies power for all of the electrical loads that are required. The battery is more effectively charged with a higher voltage when the battery is cold and a lower voltage when the battery is warm. The PCM is able to adjust the charging voltage according to the battery temperature by using a signal from the intake air temperature (IAT) sensor. This means the voltage setpoint is calculated by the PCM and communicated to the regulator by the GENCOM circuit.

The PCM simultaneously controls and monitors the output of the generator. When the current consumption is high or the battery is discharged the PCM raises engine speed to increase generator output.

To minimize the engine drag when starting the engine, the PCM does not allow the generator to produce any output until the engine has started. The PCM turns off the generator during cranking to reduce the starter load and improve cranking speed. Once the engine starts, the PCM slowly increases generator output to help establish a stable engine speed.

The PCM controls the charging system warning indicator by sending a message over the high-speed controller area network (HS-CAN) to the instrument cluster (IC). The PCM turns the charging system warning indicator off when generator output begins. The charging system warning indicator is also illuminated by the PCM whenever the key is ON with the engine OFF.

This is a System 4 charging system, which uses the GENMON and GENCOM lines to control and monitor the charging system through the PCM. System 4 charging systems are virtually identical in design and therefore, share the same diagnostics. The circuit numbers and colors may be different, but the functions are the same.

Inspection and Verification

WARNING: **Batteries contain sulfuric acid and produce explosive gases. Work in a well-ventilated area. Do not allow the battery to come in contact with flames, sparks or burning substances. Avoid contact with skin, eyes or clothing. Shield eyes when working near the battery to protect against possible splashing of acid solution. In case of acid contact with skin or eyes, flush immediately with water for a minimum of 15 minutes, then get prompt medical attention. If acid is swallowed, call a physician immediately. Failure to follow these instructions may result in serious personal injury.**

WARNING: **Always lift a plastic-cased battery with a battery carrier or with hands on opposite corners. Excessive pressure on the battery end walls may cause acid to flow through the vent caps, resulting in personal injury and/or damage to the vehicle or battery.**

CAUTION: Do not make jumper connections except as directed. Incorrect connections may damage the voltage regulator test terminals, fuses or fusible links.

CAUTION: Do not allow any metal object to come in contact with the generator housing and internal diode cooling fins. A short circuit may result and burn out the diodes.

NOTE: While carrying out any pinpoint test, disregard any DTCs set while following a specific pinpoint test. After the completion of a test, be sure to clear all DTCs in the PCM.

NOTE: All voltage measurements are referenced to the negative (-) battery post unless otherwise specified.

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Battery • Generator drive belt • Generator pulley	• Battery junction box (BJB) fuse 8 (10A) • Circuitry • Fusible links • Cables • Generator • PCM • Charging system warning indicator

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. Verify the battery condition. Refer to **BATTERY, MOUNTING & CABLES** article.
5. Check the operation of the charging system warning indicator at the instrument cluster (IC). Normal operation is as follows:
   - With the key OFF, the charging system warning indicator should be off.
   - With the key ON and the engine OFF, the charging system warning indicator should be on.
   - With the engine running, the charging system warning indicator should be off.
6. Turn off the headlamps and the A/C system (if equipped). Turn the climate control blower to low/off. Check the battery voltage before and after starting the engine to determine if the battery voltage increases.

NOTE: Make sure to use the latest scan tool software release.

7. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM.

8. If the scan tool does not communicate with the VCM:
   - check the VCM connection to the vehicle.
   - check the scan tool connection to the VCM.
   - refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
9. If the scan tool does not communicate with the vehicle:
   - verify the ignition key is in the ON position.
   - verify the scan tool operation with a known good vehicle.
   - refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
10. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
- If the network test passes, retrieve and record continuous memory DTCs.

11. Clear the continuous DTCs and carry out the self-test diagnostics for the PCM and the IC.
12. If the DTCs retrieved are related to the concern, go to the **DTC CHART** . For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
13. If no DTCs related to the concern are retrieved, go to **SYMPTOM CHART**.

NOTE: **DTC P0622 can be set by the loss of the communication lines between the generator and the PCM. The charging system warning indicator then illuminates until the engine is operated at greater than 4,500 RPM (approximately wide open throttle [WOT]) for a minimum of 3 seconds. At this time, the generator self-excites. The charging system warning indicator remains illuminated, and the generator operates in a default mode (approximately 13.5 volts) until the engine is turned off.**

DTC CHART

DTCs	Description	Source	Action
B1317	Battery Voltage High	Various Modules	Go to **Pinpoint Test B** .
B1318	Battery Voltage Low	Various Modules	Go to **Pinpoint Test F** .
B1676	Battery Voltage Out of Range	Various Modules	Go to **Pinpoint Test F** .
P0563	System Voltage High	PCM	Go to **Pinpoint Test B** .
P0620	Generator Control Circuit	PCM	Go to **Pinpoint Test B** .
P0622	Generator Field Term Circuit	PCM	Go to **Pinpoint Test B** .
P0625	Generator Field Terminal Circuit - Low	PCM	Go to **Pinpoint Test B** .
P0626	Generator Field Terminal Circuit - High	PCM	Go to **Pinpoint Test B** .
P065B	Generator Control Circuit Range/ Performance	PCM	Go to **Pinpoint Test B** .
All other DTCs	-	-	REFER to **MULTIFUNCTION ELECTRONIC MODULES** article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
• The battery is discharged	• High key-off current drain(s) • Engine, generator and battery grounds	• Go to **Pinpoint Test**

or battery voltage is low	• Positive battery cable • Circuitry • Battery • Generator	**A** .
• The charging system warning indicator is on with the engine running and no charging system DTCs present	• Circuitry • PCM • Generator • Instrument cluster (IC)	• Go to **Pinpoint Test C** .
• The generator is noisy	• Accessory drive belt • Loose bolts/brackets • Generator/pulley	• Go to **Pinpoint Test D** .
• Radio interference	• Generator • Circuitry • In-vehicle entertainment system	• Go to **Pinpoint Test E** .

Pinpoint Tests

CAUTION: Electronic modules are sensitive to electrostatic discharge. If exposed to these charges, damage can result.

Refer to **INSPECTION AND VERIFICATION** , the **DTC CHART** and the **SYMPTOM CHART** for direction to the appropriate pinpoint test.

Pinpoint Test A: The Battery is Discharged or Battery Voltage is Low

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Charging System for schematic and connector information.

Normal Operation

The generator output voltage is supplied through the positive battery output (B+), circuit SDC14 (RD) terminal on the rear of the generator to the battery and electrical system. During normal operation the charging system warning indicator is off with the key ON engine running (KOER). The charging system warning indicator is on with the key ON engine OFF (KOEO).

This pinpoint test is intended to diagnose the following:

- High key-off current drain(s)
- Engine, generator and battery grounds
- Positive battery cable
- Circuitry

- Battery
- Generator

PINPOINT TEST A: THE BATTERY IS DISCHARGED OR BATTERY VOLTAGE IS LOW

A1 CHECK THE BATTERY CONDITION

- Carry out the Battery - Condition Test to determine if the battery can hold a charge and is OK for use. Refer to **BATTERY, MOUNTING & CABLES** article.
- **Does the battery pass the condition test?**

  YES : Go to A2.

  NO : INSTALL a new battery. REFER to **BATTERY, MOUNTING & CABLES** article. TEST the system for normal operation.

A2 CHECK THE GENERATOR OUTPUT

- Carry out the Generator On-Vehicle Load Test and No Load Test. Refer to the Component Tests in this service information.
- **Does the generator pass the component tests?**

  YES : Go to A3.

  NO : Go to **Pinpoint Test B** .

A3 CHECK FOR CURRENT DRAINS

- Carry out the Battery - Drain Testing. Refer to the Component Tests in this service information.
- **Are any circuits causing excessive current drains?**

  YES : REPAIR as necessary. TEST the system for normal operation.

  NO : Go to A4.

A4 CHECK THE VEHICLE GROUNDS

- Key in START position.

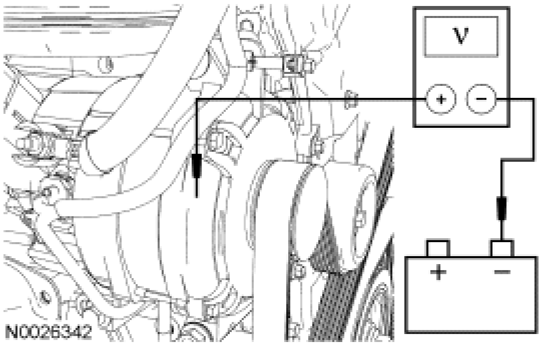


Fig. 1: Checking Vehicle Grounds
Courtesy of FORD MOTOR CO.

- With the engine running, measure the voltage drop between the generator housing and the negative battery terminal.
- **Is the voltage drop less than 0.1 volt?**

YES : Go to A5.

NO : CHECK the engine ground, generator ground and the battery ground for corrosion. TEST the system for normal operation.

A5 CHECK THE VOLTAGE DROP IN THE B+ CIRCUIT SDC14 (RD)

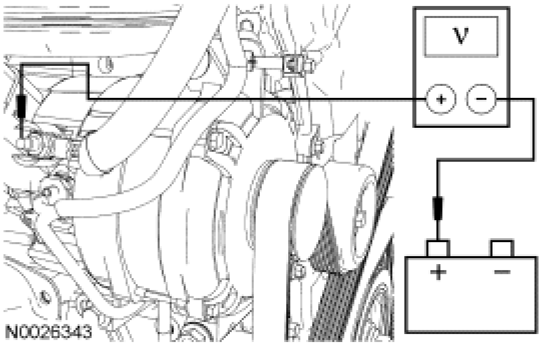


Fig. 2: Checking Voltage Drop
Courtesy of FORD MOTOR CO.

- With the engine running, measure the voltage drop between generator B+ C102b circuit SDC14 (RD) and the positive battery terminal.
- **Is the voltage drop less than 0.5 volt?**

  YES : CHECK if the customer left any electrical system(s) on or if there is an intermittent excessive battery draw. TEST the system for normal operation.

  NO : CHECK for any corrosion in the B+ SDC14 (RD), positive battery cable and/or connections. REPAIR as necessary. TEST the system for normal operation.

Pinpoint Test B: DTCs B1317, P0563, P0620, P0625, P0626 and P065B - Charging System Failure

Refer to appropriate WIRING DIAGRAMS ARTICLE, Charging System for schematic and connector information.

Normal Operation

With the engine running, the charging system warning indicator is off. The A sense circuit SBB08 (VT/RD) to the generator field coil is 13-15 volts. This voltage feedback is used by the regulator to maintain the battery voltage at the desired setpoint. The S (stator) circuit (internal to the generator) is used to monitor generator operation. This circuit and other regulator internal conditions are checked by the regulator to allow the PCM to turn off the charging system warning indicator by sending a message over the controller area network (CAN) bus to the instrument cluster (IC). The positive battery output (B+) circuit SDC14 (RD) is the generator output voltage supplied to the battery and electrical system.

This pinpoint test is intended to diagnose the following:

- Generator
- IC

- PCM

DTC Description	Fault Trigger Conditions
• B1317 - Battery Voltage High • P0563 - System Voltage High • P0620 - Generator Control Circuit • P0625 - Generator Field Circuit - Low • P0626 - Generator Field Circuit - High • P065B - Generator Control Circuit Range	Open, short to ground or battery. Charging system voltage is below 13 volts or above 16 volts and charging system lamp is illuminated.

PINPOINT TEST B: DTCs B1317, P0563, P0620, P0625, P0626 AND P065B - CHARGING SYSTEM FAILURE

NOTE: Make sure battery voltage is greater than 12.2 volts prior to carrying out this pinpoint test.

NOTE: Do not have a battery charger attached during vehicle testing.

B1 CONFIRM THE BATTERY CONDITION

- Carry out the Battery Condition Test. Refer to **BATTERY, MOUNTING & CABLES** article.
- **Is the battery OK?**

  YES : Go to B2.

  NO : CORRECT the battery condition and Go to B2.

B2 CHECK THE BATTERY JUNCTION BOX (BJB) FUSE 8 (10A)

- Check fuse: BJB 8 (10A)
- **Is BJB 8 (10A) OK?**

  YES : Go to B3.

  NO : REPAIR circuit SBB08 (VT/RD) and INSTALL a fuse. INSPECT PCM and engine ground circuits and make sure they are securely attached. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B3 CHECK THE GENERATOR B+ CONNECTION

- Key in OFF position.
- Inspect generator C102b circuit SDC14 (RD) connection. Connection should be tight.

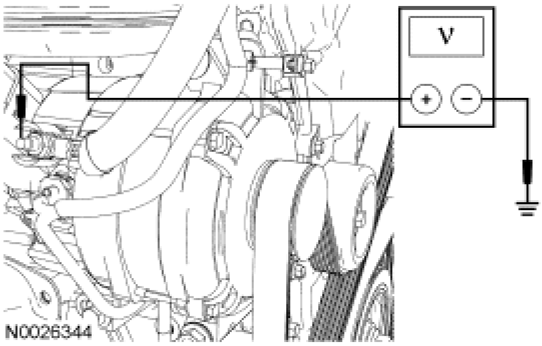


Fig. 3: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between generator C102b circuit SDC14 (RD) and ground.
- **Is generator C102b connection tight and does the generator B+ measure battery voltage?**

  YES : Go to B4.

  NO : TIGHTEN the generator B+ connection or REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B4 MONITOR THE PCM PIDs WITH KEY ON/ENGINE OFF (KOEO)

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Clear the PCM DTCs
- Enter the following diagnostic mode on the diagnostic tool: PCM DataLogger

  NOTE: Many of the PCM PIDs selected will be monitored later in this pinpoint test.

  Select and monitor the following PCM PIDs:

  - Generator Monitor (GENMON).
  - Generator Command Duty Cycle (GENCMD).
  - Generator Voltage Desired (GENVDSD).
  - Generator Fault Indicator Lamp (GENFIL).
  - Engine Revolutions Per Minute (RPM).
  - Module Supply Voltage (VPWR).
- Monitor the GENMON PID.
- **Does the PID read 0%?**

  YES : Go to B5.

  NO : Go to B8.

B5 MONITOR THE PCM PID GENMON WITH KEY ON/ENGINE RUNNING (KOER)

- Key in START position.
- With the engine at idle, wait 15 seconds for the GENVDSD PID to increase to greater than 13 volts.

- Monitor PID GENMON at idle and 3,000 RPM.
- **Does the GENMON PID read between 3% and 98% at engine idle speed and at 3,000 RPM?**

  YES : Go to B6.

  NO : Go to B8.

B6 MONITOR THE PCM PIDs GENMON, VPWR AND GENVDSD WITH THE ENGINE AT 3,000 RPM

- Turn all electrical accessories (lights, blower motor, etc.) off.

NOTE: If GENMON PID does not remain below 85%, make sure that the battery is at an acceptable state of charge and that all electrical accessories are off.

- Increase the engine speed to 3,000 RPM (or road test).
- **Does the VPWR PID remain within ± 0.5 volt of the GENVDSD PID when the GENMON PID is less than 85%?**

  YES : Go to B7.

  NO : Go to B17.

B7 MONITOR THE PCM PIDs GENMON, VPWR AND GENVDSD WITH THE ENGINE AT IDLE

- Return the engine speed to idle.

CAUTION: On vehicles with low electrical loads, it may be necessary to add external loads (devices connected to power points, etc.) to determine the maximum GENMON value. GENMON value will not read between 95%-98% on a vehicle with minimal electrical accessories. As long as there is a significant increase in the GENMON PID following the procedure below, answer YES to the question.

- Determine the maximum GENMON PID value by lowering engine idle RPM to 500 RPM or less using output state control (manual transaxle vehicles require the parking brake applied and the clutch pedal depressed) and turn on all electrical accessories until the VPWR PID is less than the GENVDSD PID by at least 0.7 volt. Under this condition the GENMON PID should read between 95% and 98%.
- **Does the GENMON PID read between 95% and 98%?**

  YES : Go to B19.

  NO : Go to B17.

B8 CHECK THE VOLTAGE OUTPUTS FROM THE PCM

- Key in OFF position.
- Disconnect: Generator C102a
- Key in ON position.

- Measure the voltage of the following circuits:

EXPECTED VOLTAGES

Generator Connector	Circuit	Expected Voltage (Approximate)
C102a-1	CDC15 (VT)	8-11 volts (should be less than battery voltage)
C102a-2	CDC10 (BU/OG)	0 volts
C102a-3	SBB08 (VT/RD)	Battery voltage

- **Are the voltages as indicated for each circuit?**

  YES : Go to B13.

  NO : If a fault is detected in the GENCOM circuit CDC10 (BU/OG) or GENMON circuit CDC15 (VT), go to B9.

  If a fault is detected in the A sense circuit SBB08 (VT/RD) circuit, go to B12.

B9 CHECK CIRCUITS CDC10 (BU/OG) AND CDC15 (VT) FOR DAMAGE OR AN OPEN

- Key in OFF position.
- Disconnect: Generator C102a
- Disconnect: PCM C175b
- Inspect the following for damaged or pushed-out pins:

Component	Connector	Circuit
PCM	C175b-22	CDC10 (BU/OG)
PCM	C175b-23	CDC15 (VT)
Generator	C102a-2	CDC10 (BU/OG)
Generator	C102a-1	CDC15 (VT)

- Measure the resistance between PCM C175b-22, circuit CDC10 (BU/OG) harness side and generator C102a-2, circuit CDC10 (BU/OG) harness side and PCM C175b-23, circuit CDC15 (VT) harness side and generator C102a-1, circuit CDC15 (VT) harness side.
- **Are the connectors and pins free of damage and are the resistances less than 5 ohms?**

  YES : Go to B10.

NO : REPAIR the affected circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B10 CHECK CIRCUITS CDC10 (BU/OG) AND CDC15 (VT) FOR SHORT TO VOLTAGE

- Key in ON position.
- Measure the voltage between generator C102a-1, circuit CDC15 (VT) harness side and ground and generator C102a-2, circuit CDC10 (BU/OG) harness side and ground.
- **Are the voltages approximately 0 volt?**

  YES : Go to B11.

  NO : REPAIR the affected circuits. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B11 CHECK CIRCUITS CDC10 (BU/OG) AND CDC15 (VT) FOR SHORT TO GROUND

- Key in OFF position.
- Measure the resistance between generator C102a-1, circuit CDC15 (VT) harness side and ground and generator C102a-2, CDC10 (BU/OG) harness side and ground and measure resistance between generator C102a-1, circuit CDC15 (VT) harness side and generator C102a-2, circuit CDC10 (BU/OG) harness side.
- **Are the resistances less than 5 ohms?**

  YES : REPAIR the affected circuits. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

  NO : Go to B24.

B12 CHECK CIRCUIT SBB08 (VT/RD) FOR DAMAGE OR AN OPEN

- Key in OFF position.
- Disconnect: Generator C102a
- Inspect C102a-3 for damaged or pushed-out pins.
- Measure the resistance of the A sense circuit SBB08 (VT/RD) between the battery and C102a-3 generator connector.
- **Are the connectors and pins free of damage and are the resistances less than 5 ohms?**

  YES : Go to B24.

  NO : REPAIR the affected circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B13 CHECK FOR SHORTED CIRCUITS

- Key in OFF position.
- Disconnect: Generator C102a
- Connect: PCM C175b
- Key in ON position.
- Verify that PID GENMON reads 100%.
- Carry out the Wiggle Test of wiring to determine if PID GENMON changes from 100%.
- **Does PID GENMON change from 100%?**

  YES : REPAIR short circuit on CDC15 (VT) C102a-1.

NO : Go to B14.

B14 CHECK THE PCM PID GENMON INPUT TO THE PCM

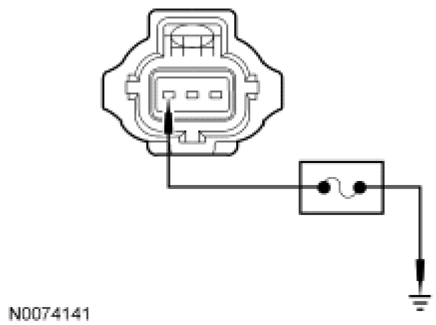


Fig. 4: Connecting Fused Jumper Wire
Courtesy of FORD MOTOR CO.

- Connect a fused (10A) jumper wire between generator C102a-1, circuit CDC15 (VT), harness side and ground.
- Key in ON position.
- Monitor the GENMON PID while performing a wiggle test on the wire harness.
- **Does the PID read 0%?**

  YES : Go to B15.

  NO : REPAIR open connection on circuit CDC15 (VT). INSPECT generator C102a-1 and PCM C175e-16 for damage. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B15 COMPARE THE PCM PIDS GENMON AND GENCMD

- Key in OFF position.
- Disconnect: Fused (10A) Jumper Wire

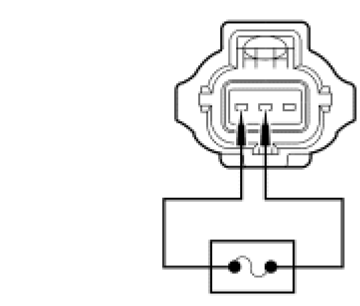


Fig. 5: Connecting Fused Jumper Wire
Courtesy of FORD MOTOR CO.

- Connect a fused (10A) jumper wire between generator C102a-1, circuit CDC15 (VT), harness side and generator C102a-2, circuit CDC10 (BU/OG), harness side.

- Key in ON position.
- Monitor the GENMON and GENCMD PIDs while performing a wiggle test on the harness.
- **Does the GENMON PID read within 2% of the GENCMD PID?**

  YES : Go to B16.

  NO : INSPECT C175b-66, circuit GD124 (BK/VT), PCM ground circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B16 A SENSE CIRCUIT LOAD TEST

NOTE: This step puts a load on the A sense circuit. If there are corroded or loose connections, loading the circuit may help show the fault.

- Key in ON position.
- Using a 12-volt test lamp, check for voltage at C102a-3, circuit SBB08 (VT/RD).
- **Does the test lamp illuminate?**

  YES : Go to B23.

  NO : REPAIR connection on circuit SBB08 (VT/RD). INSPECT generator C102a-3 for damage. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B17 CHECK THE PCM VPWR PID

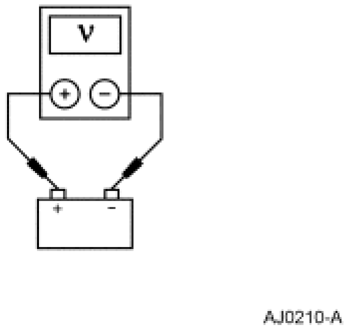


Fig. 6: Checking Charging System Voltage
Courtesy of FORD MOTOR CO.

- With the engine running, measure the battery voltage.
- Monitor the VPWR PID.
- **Are the battery voltage and PID within 0.5 volt of each other?**

  YES : Go to B19.

  NO : Go to B18.

B18 MEASURE THE PCM INPUT VOLTAGE

- Key in OFF position.
- Disconnect: PCM C175b
- Connect a fused (10A) jumper wire between C175b-8 and ground.

- Key in ON position.

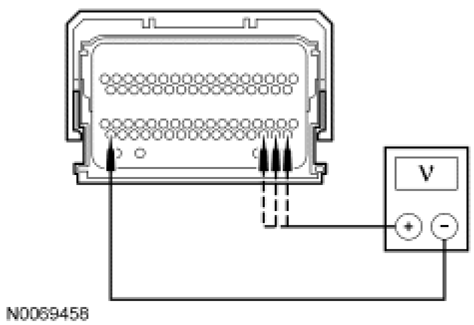


Fig. 7: Checking Circuits For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between PCM input voltage pins 51, 52 and 53 to PCM ground pin 66.
- **Are voltages within 0.5 volt of PID VPWR?**

  YES : Go to B24.

  NO : REPAIR high resistance or loose connections between C175b-51 circuit CBB30 (YE/BU), C175b-52 circuit CBB30 (YE/BU), C175b-53 circuit CBB30 (YE/BU) PCM power circuits or C175b-10, circuit GD122 (BK), PCM ground circuit.

B19 CHECK THE GENERATOR B+ RESISTANCE

- Key in OFF position.
- Disconnect: Battery
- Disconnect: Generator C102b

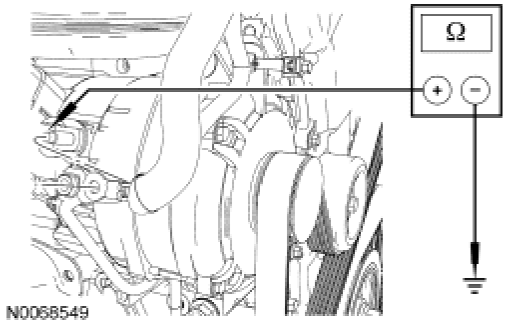


Fig. 8: Checking Circuit For Short To Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between generator C102b, component side and the generator housing.
- **Is the resistance greater than 125K ohms?**

  YES : Go to B20.

  NO : INSTALL a new generator. REFER to **GENERATOR & REGULATOR** article. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B20 CHECK THE RESISTANCE OF THE VOLTAGE REGULATOR INTERNAL CIRCUITS TO GROUND

- Measure the resistance between generator C102a, component side and ground. Refer to the following table.

Pin	Expected Resistance
1	Greater than 1,000K ohms
2	Greater than 125K ohms
3	Greater than 125K ohms

- **Are the resistance values as indicated?**

  YES : Go to B21.

  NO : INSTALL a new generator. REFER to **GENERATOR & REGULATOR** article. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B21 ROAD TEST IN AN ATTEMPT TO VERIFY THE FAULT

- Connect: Generator C102b
- Connect: Generator C102a
- Connect: Battery
- Carry out a road test and monitor the following PIDs:
  - Generator monitor (GENMON).
  - Generator voltage desired (GENVDSD).
  - Module supply voltage (VPWR).
- **Does the GENMON PID read either 0% or 100% during the road test?**

  YES : Go to B2.

  NO : If a road test did not cause the fault to occur, intermittent fault condition may be present. Go to B22.

B22 USE THE VEHICLE DATA RECORDER (VDR) TO CAPTURE INTERMITTENT FAULT CONDITION

- Connect a VDR to the vehicle and setup to capture the following PIDs:
  - Generator monitor (GENMON).
  - Generator command duty cycle (GENCMD).
  - Generator voltage desired (GENVDSD).
  - Generator fault indicator lamp (GENFIL).

    - Engine revolutions per minute (RPM).
    - Module supply voltage (VPWR).
- Set the VDR to trigger if any of the following events occur during vehicle operation (waiting 15 seconds after start):
    - VPWR PID reads greater than 15.2 volts.
    - GENMON PID reads 0% or 100%.
- Road test the vehicle.
- **Was a fault captured by the VDR?**

  YES : RECORD any values from the VDR that may have been captured. If no charging system DTCs are present, go to **Pinpoint Test C** . If any charging system DTCs are present, go to B3.

  NO : No problem found at this time. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B23 CHECK THE CHARGING SYSTEM CIRCUITS FOR INTERMITTENT FAULTS

- Key in OFF position.
- Connect: Generator C102b
- Connect: Generator C102a
- Connect: PCM C175b
- Connect the diagnostic tool.
- Key in START position.
- With the engine running, monitor the charging system warning indicator lamp and the scan tool for DTCs.
- **Does the charging system warning indicator lamp illuminate and does any charging system DTC get stored into memory?**

  YES : INSTALL a new generator. REFER to **GENERATOR & REGULATOR** article. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

  NO : The fault is not present and cannot be recreated at this time. This may indicate an intermittent fault. Go to B19.

B24 CHECK THE CHARGING SYSTEM CIRCUITS FOR INTERMITTENT FAULTS

- Key in OFF position.
- Connect: Generator C102b
- Connect: Generator C102a
- Connect: PCM C175b
- Connect the diagnostic tool.
- Key in START position.
- With the engine running, monitor the charging system warning indicator lamp and the scan tool for DTCs.
- **Does the charging system warning indicator lamp illuminate and does any charging system DTC get stored into memory?**

  YES : INSTALL a new PCM. REFER to **ELECTRONIC ENGINE CONTROLS** article.

NO : The fault is not present and cannot be recreated at this time. This may indicate an intermittent fault. Go to B19.

Pinpoint Test C: The Charging System Warning Indicator is On With the Engine Running and No Charging System DTCs Present

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Charging System for schematic and connector information.

Normal Operation

With the engine running, the charging system warning indicator is off. The A sense circuit SBB08 (VT/RD) from the battery B+ to the generator regulator is 13-15 volts. This voltage feedback is used by the regulator to maintain the battery voltage at the desired setpoint. The S (stator) circuit (internal to the generator) is used to monitor generator operation. This circuit and other regulator internal conditions are checked by the regulator to allow the PCM to turn off the charging system warning indicator by sending a message over the controller area network (CAN) bus to the instrument cluster (IC). The positive battery output (B+) circuit SDC14 (RD) is the generator output supplied to the battery and electrical system.

This pinpoint test is intended to diagnose the following:

- Generator
- Circuitry
- IC
- PCM

PINPOINT TEST C: THE CHARGING SYSTEM WARNING INDICATOR IS ON WITH THE ENGINE RUNNING AND NO CHARGING SYSTEM DTCs PRESENT

NOTE:	**Make sure battery voltage is greater than 12.2 volts prior to carrying out this pinpoint test.**

NOTE:	**Do not have a battery charger attached during vehicle testing.**

C1 CHECK THE BATTERY CONDITION

- Carry out the Battery - Condition Test to determine if the battery can hold a charge and is OK for use. Refer to **BATTERY, MOUNTING & CABLES** article.
- **Does the battery pass the condition test?**

  YES : Go to C2.

  NO : INSTALL a new battery. REFER to **BATTERY, MOUNTING & CABLES** article. TEST the system for normal operation.

C2 CHECK THE GENERATOR B+ CONNECTION

- Key in OFF position.

- Inspect generator C102b circuit SDC14 (RD) connection. Connection should be tight.

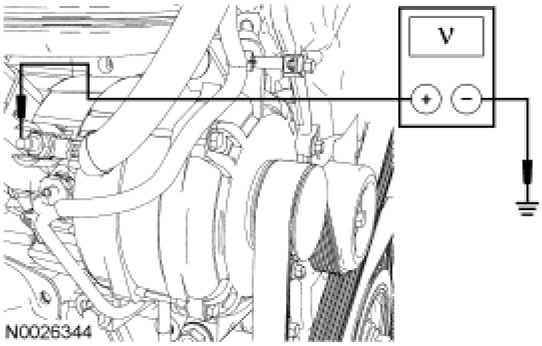


Fig. 9: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between generator C102b circuit SDC14 (RD) and ground.
- **Is generator C102b connection tight and does the generator B+ measure battery voltage?**

  YES : Go to C3.

  NO : TIGHTEN the generator B+ connection or REPAIR the circuit. CLEAR the any DTCs. REPEAT the self-test. TEST the system for normal operation.

C3 CHECK THE VOLTAGE DROP IN THE B+ CIRCUIT SDC14 (RD)

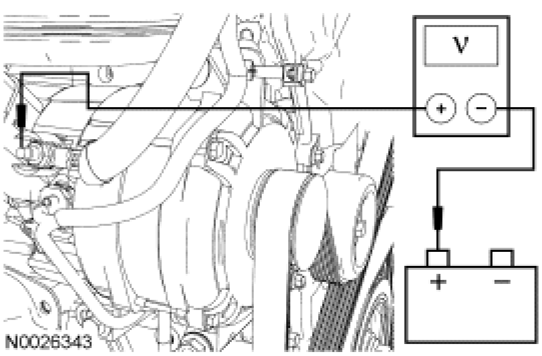


Fig. 10: Checking Voltage Drop
Courtesy of FORD MOTOR CO.

- With the engine running, measure the voltage drop between generator B+ C102b circuit SDC14 (RD) and the positive battery terminal.
- **Is the voltage drop less than 0.5 volt?**

  YES : Go to C4.

  NO : CHECK for any corrosion in the B+ C102b circuit SDC14 (RD), positive battery cable and/or connections. REPAIR as necessary. TEST the system for normal operation.

C4 CHECK THE DTCs IN THE PCM

- Key in OFF position.
- Connect the diagnostic tool.
- Key in START position.

- Enter the following diagnostic mode on the diagnostic tool: Retrieve PCM DTCs
- Use the recorded PCM DTCs from the continuous and on-demand self tests.
- **Are any PCM DTCs recorded?**

  YES : For all DTCs EXCEPT charging system DTCs, REFER to the **INTRODUCTION - GASOLINE ENGINES** article. If any charging system DTCs are present, go to **Pinpoint Test B** . If referred here by the PC/ED, go to C5.

  NO : Go to C5.

C5 MONITOR PID GENFIL

- With the engine running, monitor PID GENFIL.
- **Does PID GENFIL show fault?**

  YES : Go to C6.

  NO : REPAIR/INSTALL a new IC, IC-to-PCM connection or the PCM. TEST the system for normal operation.

C6 MONITOR PID VPWR

- With the engine running, monitor PIDS VPWR, GENFIL, GENVDSD.
- **Is PID VPWR greater than 15.6 volts?**

  YES : Go to C7.

  NO : Go to C8.

C7 CHECK PCM VPWR PID

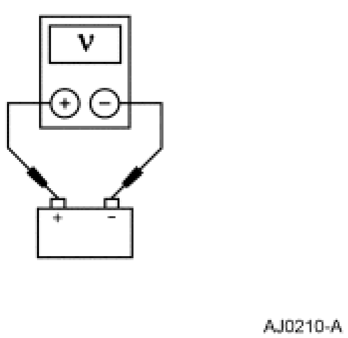


Fig. 11: Checking Charging System Voltage
Courtesy of FORD MOTOR CO.

- With the engine running, measure the battery voltage.
- Monitor the VPWR PID.
- **Are the battery voltage and VPWR PID within 0.6 volt of each other?**

  YES : Go to C8.

  NO : Go to C9.

C8 MEASURE THE PCM INPUT VOLTAGE

- Key in OFF position.
- Disconnect: PCM C175b

- Connect a fused (10A) jumper wire between C175b-8 and ground.
- Key in ON position.

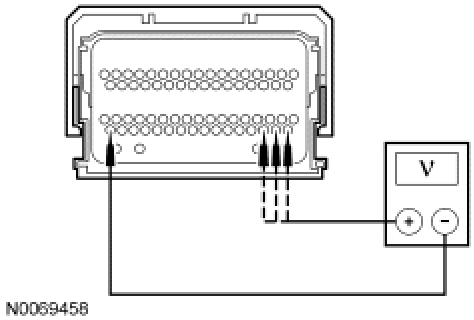


Fig. 12: Checking Circuits For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between PCM input voltage pins 51, 52 and 53 to PCM ground pin 66.
- **Are voltages within 0.5 volt of PID VPWR?**

  YES : INSTALL a new PCM.

  NO : REPAIR high resistance or loose connections between C175b-51 circuit CBB30 (YE/BU), C175b-52 circuit CBB30 (YE/BU), C175b-53 circuit CBB30 (YE/BU) PCM power circuits or C175b-10, circuit GD122 (BK), PCM ground circuit.

C9 MONITOR THE PCM PID GENERATOR MONITOR WITH KEY ON/ENGINE RUNNING

- Key in START position.
- With the engine at idle, wait 15 seconds for the GENVDSD PID to increase to greater than 13 volts.
- Monitor PID GENMON at idle and 3,000 RPM.
- **Does the GENMON PID read between 3% and 98% at engine idle speed and at 3,000 RPM?**

  YES : Go to C10.

  NO : Go to C11.

C10 MONITOR PID GENVDSD

- With the engine running, monitor PID GENVDSD.
- **Is PID GENVDSD greater than 15.2 volts?**

  YES : REPAIR connection on circuit SBB08 (VT/RD). INSPECT generator C102a-3 for damage. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation. If no problems are found, INSTALL a new generator. REFER to **GENERATOR & REGULATOR** article. TEST the system for normal operation.

  NO : INSTALL a new PCM.

C11 CHECK FOR SHORTED CIRCUITS

- Key in OFF position.
- Disconnect: Generator C102a
- Verify that PID GENMON reads 100%.

- Carry out the Wiggle Test of wiring to determine if PID GENMON changes from 100%.
- **Does PID GENMON change from 100%?**

  YES : REPAIR short circuit on C102a-1 CDC15 (VT).

  NO : Go to C12.

C12 CHECK THE PCM PID GENERATOR MONITOR INPUT TO THE PCM

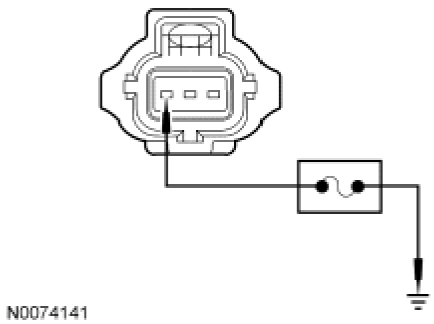


Fig. 13: Connecting Fused Jumper Wire
Courtesy of FORD MOTOR CO.

- Connect a fused (10A) jumper wire between generator C102a-1, circuit CDC15 (VT), harness side and ground.
- Key in ON position.
- Monitor the GENMON PID while carrying out a Wiggle Test on the wire harness.
- **Does the GENMON PID read 0%?**

  YES : REPAIR open connection on circuit CDC15 (VT) C102a-1 and PCM C175b-23. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

  NO : INSTALL a new generator. REFER to **GENERATOR & REGULATOR** article. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test D: The Generator is Noisy

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Charging System for schematic and connector information.

Normal Operation

The generator is belt-driven by the engine accessory drive system.

This pinpoint test is intended to diagnose the following:

- Accessory drive belt
- Loose bolts/brackets
- Generator/pulley

PINPOINT TEST D: THE GENERATOR IS NOISY

D1 CHECK FOR ACCESSORY DRIVE BELT NOISE AND LOOSE MOUNTING BRACKETS

- Key in OFF position.
- Check the accessory drive belt for damage and correct installation. Refer to **ACCESSORY DRIVE** article.
- Check the accessory mounting brackets and generator pulley for looseness or misalignment.
- **Is the accessory drive OK?**

  YES : Go to D2.

  NO : REPAIR as necessary. REFER to **ACCESSORY DRIVE** article for diagnosis and testing of the accessory drive system. TEST the system for normal operation.

D2 CHECK THE GENERATOR MOUNTING

- Check the generator mounting for loose bolts or misalignment.
- **Is the generator mounted correctly?**

  YES : Go to D3.

  NO : REPAIR as necessary. TEST the system for normal operation.

D3 CHECK THE GENERATOR FOR ELECTRICAL NOISE

- Disconnect: Generator C102b
- Key in START position.
- With the engine running, use a stethoscope or equivalent listening device to probe the generator.
- **Is the noise still present?**

  YES : Go to D4.

  NO : INSTALL a new generator. REFER to **GENERATOR & REGULATOR** article. TEST the system for normal operation.

D4 CHECK THE GENERATOR FOR MECHANICAL NOISE

- With the engine running, use a stethoscope or equivalent listening device to probe the generator and the accessory drive area for unusual mechanical noise.
- **Is the generator the noise source?**

  YES : INSTALL a new generator. REFER to **GENERATOR & REGULATOR** article. TEST the system for normal operation.

  NO : REFER to **ENGINE SYSTEM - GENERAL INFORMATION** article to diagnose the source of the engine noise.

Pinpoint Test E: Radio Interference

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Charging System for schematic and connector information.

Normal Operation

The generator radio suppression equipment reduces interference transmitted through the speakers by the vehicle electrical system.

This pinpoint test is intended to diagnose the following:

- Generator
- Circuitry
- In-vehicle entertainment system

PINPOINT TEST E: RADIO INTERFERENCE

E1 VERIFY THE GENERATOR IS THE SOURCE OF THE RADIO INTERFERENCE

NOTE: **If the OEM audio unit has been replaced with an aftermarket unit, the vehicle may not pass this test. Return the vehicle to OEM condition before following this pinpoint test.**

- Key in START position.
- Start and run the engine.
- Tune the audio unit to a station where the interference is present.
- Key in OFF position.
- Disconnect: Generator C102b
- Key in START position.
- With the engine running, determine if the interference is still present.
- **Is the interference present with the generator disconnected?**

  YES : REFER to **INFORMATION & ENTERTAINMENT SYSTEMS** article for diagnosis and testing of the in-vehicle entertainment system.

  NO : INSTALL a new generator. REFER to **GENERATOR & REGULATOR** article. TEST the system for normal operation.

Pinpoint Test F: DTC B1318 or B1676 - Battery Voltage Low or Battery Voltage Out of Range

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Charging System for schematic and connector information.

Normal Operation

Various control modules within the vehicle operate at a voltage between 10-17 volts. Voltage to these modules is supplied through the vehicle harness to the module(s). Ground for the modules is provided by the modules ground circuit(s). If voltages to these modules rise above 17 volts, DTC B1317 or B1676 may set or if voltage drops below 10 volts, DTC B1318 or B1676 may set.

- DTC B1318 Battery Voltage Out Of Range - If charging system voltage falls below a certain voltage, some modules within the vehicle may set this DTC. Some modules have a different voltage minimum set point to set this DTC.

- DTC B1676 Battery Voltage High - If charging system voltage falls below or above a certain voltage, some modules within the vehicle may set this DTC. Some modules have a different voltage minimum or maximum set point to set this DTC.

This pinpoint test is intended to diagnose the following:

- Battery
- Charging system
- Module voltage supply
- Module ground

NOTE: DTC B1317 or B1318 can be set if the vehicle has had a discharged battery, has been recently jump started or has had the vehicle battery charged.

PINPOINT TEST F: DTC B1318 OR B1676 - BATTERY VOLTAGE LOW OR BATTERY VOLTAGE OUT OF RANGE

F1 CHECK FOR DTCs

- Key in OFF position.
- Connect the diagnostic tool.
- Enter the following diagnostic mode on the diagnostic tool: Retrieve all continuous DTCs
- **Is DTC B1318 or B1676 present in only one module?**

  YES : If DTC B1317 or B1676 are recorded in only one module, go to F2.

  NO : If DTC B1318 or B1676 are recorded in more than one module, go to **Pinpoint Test B** .

F2 CHECK THE BATTERY VOLTAGE

- Measure the battery voltage between the positive and negative battery posts with the key ON engine OFF (KOEO), and with the engine running, all accessory loads OFF.
- **Is the battery voltage between 10 and 13 volts with KOEO, and between 11 and 17 volts with the engine running?**

  YES : Go to F3.

  NO : CHECK and/or REPAIR the charging system as necessary. Go to **Pinpoint Test B** . TEST the system for normal operation. CLEAR the DTCs. REPEAT the self-test.

F3 CHECK THE VOLTAGE TO THE MODULE

- Key in OFF position.
- Disconnect: Affected module which recorded the DTC B1317 or B1676
- Key in ON position.
- Measure all of the affected module's power circuits, harness side to ground. Refer to affected module Wiring Diagrams for schematic and connector information. Refer to appropriate SYSTEM WIRING DIAGRAMS article.
- **Is the voltage greater than 10 volts?**

  YES : Go to F4.

NO : REPAIR the module power circuit(s). CLEAR the DTCs. REPEAT the self-test.

F4 CHECK THE MODULE GROUNDS

- Key in OFF position.
- Measure the resistance between all of the affected module's ground circuits, harness side to ground. Refer to affected module Wiring Diagrams for schematic and connector information. Refer to appropriate SYSTEM WIRING DIAGRAMS article.
- **Is the resistance less than 5 ohms?**

  YES : Go to F5.

  NO : REPAIR the module ground circuit(s). CLEAR the DTCs. REPEAT the self-test.

F5 CHECK THE MODULE FOR CORRECT OPERATION

- Key in OFF position.
- Disconnect: All of the affected module connectors
- Check the connector and module for:
  - corrosion.
  - pushed out or damaged pins.
  - connector or module damage.
- Connect: All of the affected module connectors
- Clear all DTCs.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

  YES : INSTALL a new module. REFER to the removal and installation procedure in the affected module's service information . CLEAR the DTCs. REPEAT the self-test.

  NO : The system is operating correctly at this time. The concern may have been the result of a loose connection, may recently have had a discharged battery or may have been jump started. CLEAR the DTCs. REPEAT the self-test.

Component Tests

Battery - Drain Testing

WARNING: **Do not attempt to service, charge, recharge or jump start a battery unless the following information has been read and understood. Failure to follow this instruction may result in serious personal injury.**

CAUTION: To prevent damage to the meter, do not crank the engine or operate accessories that draw more than 10A.

CAUTION: If equipped with the CD6 audio unit, precautions must be taken when the battery has been disconnected. When reconnecting the battery, make sure no interruption of power occurs for 30 seconds. If power is interrupted

during the first 30 seconds, permanent damage to the CD6 audio unit will result.

NOTE: **No factory-equipped vehicle should have more than a 50 mA (0.050 amp) draw.**

Check for current drains on the battery in excess of 50 mA (0.050 amp) with all the electrical accessories off and the vehicle at rest for at least 40 minutes. Current drains can be tested with the following procedure.

NOTE: **Many electronic modules draw 10 mA (0.010 amp) or more continuously.**

NOTE: **Use an in-line ammeter between the negative battery post and its respective cable.**

NOTE: **Typically, a drain of approximately one amp is attributed to an engine compartment lamp, glove compartment lamp or interior lamp staying on continually. Other component failures or wiring shorts are located by selectively pulling fuses to pinpoint the location of the current drain. When the current drain is found, the meter reading falls to an acceptable level. If the drain is still not located after checking all the fuses, it is due to the generator.**

NOTE: **To accurately test the drain on a battery, an in-line ammeter must be used. Use of a test lamp or voltmeter is not an accurate method due to the number of electronic modules.**

1. Make sure the junction box(es)/fuse panel(s) is accessible without turning on the interior lights or the underhood lights.
2. Drive the vehicle at least 5 minutes and over 48 km/h (30 mph) to turn on and activate the vehicle systems.
3. Allow the vehicle to sit with the key off for at least 40 minutes to allow the modules to time out/power down.
4. Connect a fused jumper wire (30A) between the negative battery cable and the negative battery post to prevent modules from resetting and to catch capacitive drains.
5. Disconnect the negative battery cable from the negative battery post without breaking the connection of the jumper wire.

   NOTE: **It is very important that continuity is not broken between the battery and the negative battery cable when connecting the meter. If this happens, the entire procedure must be repeated.**

6. Connect the battery tester between the negative battery cable and the post. The meter must be capable of reading milliamps and should have a 10 amp capability.

   NOTE: **If the meter settings need to be switched or the test leads need to be**

moved to another jack, the jumper wire must be reinstalled to avoid breaking continuity.

NOTE: **Amperage draw varies from vehicle to vehicle depending on the equipment package. Compare to a similar vehicle for reference.**

NOTE: **No factory-equipped vehicle should have more than a 50 mA (0.050 amp) draw.**

7. Remove the jumper wire.
8. Note the amperage draw. Draw varies from vehicle to vehicle depending on the equipment package. Compare to a similar vehicle for reference.
9. If the draw is found to be excessive, remove the fuses from the smart junction box (SJB) one at a time and note the current reading. Do not reinstall the fuses until you have finished testing. To correctly isolate each of the circuits, all of the fuses may need to be removed and then install one fuse, note the amperage draw, remove the fuse and install the next fuse. Continue this process with each fuse.
10. If the current draw is still excessive, remove the fuses from the battery junction box (BJB) one at a time and note the current drop. Do not reinstall the fuses until you have finished testing. To correctly isolate each of the circuits, all of the fuses may need to be removed, then install one fuse, note the amperage draw, remove the fuse and install the next fuse. Continue this process with each fuse. When the current level drops to an acceptable level after removing a fuse, the circuit containing the excessive draw has been located.
11. Check the wiring diagrams for any circuits that run from the battery without passing through the BJB or the SJB. Refer to appropriate SYSTEM WIRING DIAGRAMS article. If the current draw is still excessive, disconnect these circuits until the draw is found. Disconnect the generator electrical connections if the draw cannot be located. The generator may be internally shorted, causing the current drain.

Generator On-Vehicle Tests

CAUTION: **To prevent damage to the generator, do not make the jumper wire connections except as directed.**

CAUTION: **Do not allow any metal object to come in contact with the housing and the internal diode cooling fins with the key on or off. A short circuit may result and burn out the diodes.**

NOTE: **Battery posts and cable clamps must be clean and tight for accurate meter indications.**

NOTE: **Refer to the battery tester manual for complete directions for testing the charging system.**

1. Turn off all lamps and electrical components.
2. Place the transmission in NEUTRAL and apply the parking brake.
3. Carry out the Load Test and No-Load Test according to the following component tests:

Generator On-Vehicle Tests - Load Test

1. Switch the tester to the ammeter function.
2. Connect the positive and negative leads of the tester to the corresponding battery terminals.
3. Connect the current probe to the generator B+ output terminal, circuit SDC14 (RD).
4. With the engine running at approximately 2,000 RPM, adjust the tester load bank to determine the output of the generator. Generator output should be greater than the graph shown below. If not, refer back to the pinpoint test or Go to **Symptom Chart**.

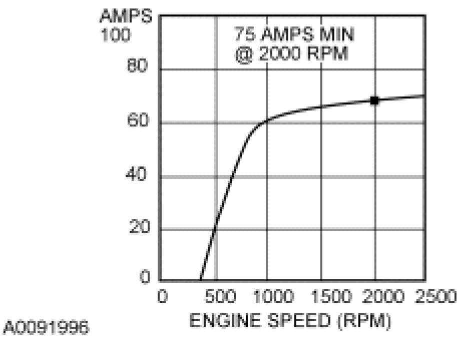


Fig. 14: Load Test Output Chart
Courtesy of FORD MOTOR CO.

Generator On-Vehicle Tests - No-Load Test

1. Switch the tester to the voltmeter function.
2. Connect the voltmeter positive lead to the generator B+ terminal, circuit SDC14 (RD), and the negative lead to ground.
3. Turn all electrical accessories off.
4. With the engine running at approximately 2,000 RPM, check the generator output voltage. The voltage should be between 13.2 and 15.5 volts. If not, refer back to the pinpoint test or Go to **Symptom Chart**.